Please show as much of your work as possible! Answers that are not legible will not be given credit. Write answers in the spaces provided.

Name:____

1. **(3 points)** Design a feedforward neural network to compute the XNOR function:

x_1	x_2	$f(x_1, x_2)$
0	0	1
0	1	-1
1	0	-1
1	1	1

Use a single output node that uses the Sign activation function $f(x) = \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x \leq 0 \end{cases}$ Use a single hidden layer with ReLU activation functions. Describe all weights and offsets (bias terms).

2. **(1 point)** What problem do LSTMs address that vanilla recurrent neural networks (e.g., Elman Networks) suffer from? (select the best answer from the choices below)

   (a) Elman Networks suffer from slow training times

   (b) Elman Networks suffer from the problem of vanishing gradients

   (c) LSTMs have fewer parameters, so they are likely to overfit

   (d) Elman Networks are more difficult to parallelize

   (e) Elman Networks have lower accuracy

Cont.

3. **(1 point)** Which of the following best describes the BLEU score (select the best answer below):
   (a) The objective function used for training neural machine translation systems.
   (b) A metric that is used to evaluate translation quality in which human judges rate translations on a scale from 1-5.
   (c) A metric that is used to automatically evaluate machine translation systems.
   (d) A way to measure the difficulty of translation between a given language pair (e.g., English/French).

4. **(2 points)** Manually annotate the following sentence with named entities (Person, Location, Organization), using the BIO encoding scheme, which is commonly used to formulate named entity recognition as a sequence labeling problem.

   Barack Obama will travel to Hangzhou today.

5. **(2 points)** In PyTorch, what is the purpose of calling `model.eval()` and `with torch.no_grad()` before evaluating a model's performance on a validation or test set? Explain what each does individually to highlight the differences between them.

6. **(1 points)** In the context of the auto-regressive encoder-decoder LSTMs (without attention) that we discussed in class, select all the true statements:

   (a) The encoder and the decoder need to have the same architecture.

   (b) The encoder consumes a sequence of tokens and produces a vector that is used as the initial hidden state for the decoder.

   (c) The decoder takes in two inputs: the hidden state and the predicted token.

   (d) Decoding can be efficiently parallelized across the output tokens using a GPU.

   (e) None of the above.

7. Which of the following statements about word embeddings is true? Select all that apply.

   (a) A one-hot encoding is a type of dense word embedding (e.g., is not a sparse vector).

   (b) Word2Vec and GloVe embeddings assign different vectors to the same word depending on its surrounding context.

   (c) Pre-trained embeddings from Word2Vec and GloVe cannot be updated during fine-tuning.

   (d) Contextualized embeddings (e.g., from BERT) can represent the same word differently depending on the surrounding words.

   (e) None of the above.

The End.